Check for updates

Received: 27 November 2024 | Revised: 25 February 2025 | Accepted: 12 March 2025

DOI: 10.1096/fj.202403009R

RESEARCH ARTICLE

The FASEB Journal

Potential of facial biomarkers for Alzheimer's disease and obstructive sleep apnea in Down syndrome and general population

L. M. Echeverry-Quiceno[1] | S. Llambrich[1] | Á. Heredia-Lidón[2] | S. Giménez[3,4,5] | M. Rozalem-Aranha[4,5,6] | P. Inampudi[7] | Y. Heuzé[8] | X. Sevillano[2] | J. Fortea[4,5] | N. Martínez-Abadías[1] | for the Alzheimer's Disease Neuroimaging Initiative

[1]Departament de Biologia Evolutiva, Ecologia i Ciències Ambientals (BEECA), Facultat de Biologia, Universitat de Barcelona (UB), Barcelona, Spain

[2]HER - Human-Environment Research Group, La Salle - Universitat Ramon Llull, Barcelona, Spain

[3]Multidisciplinary Sleep Unit - Respiratory Department, Hospital de la Santa Creu i Sant Pau, Biomedical Research Institute Sant Pau, Institut de Recerca Sant Pau -, Universitat Autònoma de Barcelona, Barcelona, Spain

[4]Memory Unit, Department of Neurology, Institut de Recerca Sant Pau - Hospital de la Santa Creu i Sant Pau, Universitat Autònoma de Barcelona, Madrid, Spain

[5]Centro de Investigación Biomédica en Red en Enfermedades Neurodegenerativas (CIBERNED), Madrid, Spain

[6]Neuroradiology Section, Department of Radiology, Hospital de la Santa Creu i Sant Pau, Universitat Autònoma de Barcelona, Barcelona, Spain

[7]Columbia University College of Dental Medicine, New York, USA

[8]Univ. Bordeaux, CNRS, Ministère de la Culture, PACEA, UMR 5199, Pessac, France

Correspondence
N. Martínez-Abadías, Departament de Biologia Evolutiva, Ecologia i Ciències Ambientals (BEECA), Facultat de Biologia, Universitat de Barcelona (UB), Avinguda Diagonal, 643, Les Corts, 08028 Barcelona, España.
Email: neusmartinez@ub.edu

Funding information
Fondation Jérôme Lejeune, Grant/Award Number: 2020b cycle-Project 2001, JLF1801 and No.55(2023Edition); Generalitat de Catalunya, Grant/

Abstract

Down syndrome (DS), caused by trisomy 21, is associated with an increased risk of Alzheimer's disease (AD) and obstructive sleep apnea (OSA). Traditional diagnostic methods for AD and OSA, like cerebrospinal fluid analysis and polysomnography, are invasive and challenging for people with DS. In this study, we assessed whether facial morphology could be used as a potential noninvasive biomarker for these conditions in both DS and the general population. We performed a comprehensive 3D analysis of facial shape variation by registering the 3D coordinates of 21 landmarks on facial models extracted from magnetic resonance images of 131 individuals with DS and 216 euploid (EU) adult controls, including AD and OSA

Abbreviations: AD, Alzheimer's disease; ADNI, Alzheimer's Disease Neuroimaging Initiative; AHI, Apnea–Hypopnea Index; APP, Amyloid Precursor Protein; Aβ, β-amyloid; BMD, bone mineral density; CS, centroid size; CSF, cerebrospinal fluid; CVA, canonical variate analyses; DS, Down syndrome; EU, euploid control; GM, geometric morphometrics; GPA, generalized procrustes analysis; MRI, magnetic resonance image; MV-CNN, Multi-View Consensus Convolutional Neural Networks; NFL, neurofilament light; OSA, obstructive sleep apnea; PSG, polysomnography; RARC, Resource Allocation Review Committee.

Data used in preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). As such, the investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data but did not participate in analysis or writing of this report. A complete listing of ADNI investigators can be found at: adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf.

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited and is not used for commercial purposes.
© 2025 The Author(s). *The FASEB Journal* published by Wiley Periodicals LLC on behalf of Federation of American Societies for Experimental Biology.

Award Number: 2024 FI-3 00160 and 2023 FI-2 00160; MEC | Instituto de Salud Carlos III (ISCIII), Grant/Award Number: PI14/01126, PI17/01019 and PI20/00836; HHS | National Institutes of Health (NIH), Grant/Award Number: 1R01AG056850-01A1, R21AG056974, R01AG061566 and 1RF1AG080769-01; Alzheimer's Association (AA) and Global Brain Health Institute (GBHI), Grant/Award Number: UK-23-971107; Agència de Gestió d'Ajuts Universitaris i de Recerca (AGAUR), Grant/Award Number: 2021 SGR00706 and 2021SGR01396; MEC | Agencia Española de Investigación (AEI), Grant/Award Number: PID2020-113609RB-C21/AEI/10.13039/501100011033; Fundación Álvaro Entrecanales-Lejeune

cases. Procrustes ANOVA and MANOVA quantified shape variation by sex, age, and facial size, while geometric morphometrics assessed diagnostic group differences. Significant facial shape differences were observed between the DS and EU groups, indicating sex-dependent differences and altered age-related changes in DS, particularly in females. Facial shape correlated with the amyloid beta ratio (Aβ1-42/Aβ1-40), a key AD biomarker. In DS, facial shape differences by AD diagnosis were not significant after adjusting for age and facial size, but significant shape differences were detected in the EU population. For OSA, facial shape correlated with the apnea–hypopnea index (AHI), and DS individuals with severe OSA showed distinct facial morphology compared with those without OSA, suggesting an association between facial shape and sleep respiratory disturbances. These results highlight the potential of facial morphology as a noninvasive biomarker for AD and OSA detection and management.

KEYWORDS

dementia, facial development, facial dysmorphology, geometric morphometrics, ontogenetic trajectories, sexual dimorphism, sleep and respiratory disorders, trisomy 21

1 | INTRODUCTION

Down syndrome (DS) (OMIM 190685) is a complex genetic disorder occurring in one out of 700–1000 live births that is caused by trisomy of chromosome 21.[1] The overexpression of more than 200 triplicated genes leads to a dosage imbalance, disrupting signaling pathways that regulate the development of multiple systems, including the nervous, cardiac, immune, gastrointestinal, and skeletal systems.[2] Advances in research, early intervention, educational therapy, and improvements in medical care have facilitated the management of pathologies associated with DS that manifest early in life. As a result, the quality of life of patients with DS has significantly improved, with life expectancy increasing from 9 to 60 years.[1,3,4] However, this increased longevity has led to greater vulnerability to age-related comorbidities,[1,5] such as Alzheimer's disease (AD)[6] and obstructive sleep apnea (OSA).[7] These disorders further impact cognitive impairment and substantially reduce the quality of life in people with DS.

Individuals diagnosed with DS present an ultrahigh risk of developing early-onset AD[6] In DS, the triplication of the *Amyloid Precursor Protein* (*APP*) gene, located in Hsa21,[8] is considered a necessary and sufficient condition for the overproduction of β-amyloid peptides (Aβ) in the brain and the development of AD.[8] Furthermore, individuals with DS also present a higher prevalence of severe sleep disruptions (74.5%) and OSA (78%).[7,9] Typical features associated with DS, such as midfacial flatness and mandibular hypoplasia,[10,11] muscular hypotonia,[10,11] adenotonsillar hypertrophy, and macroglossia,[12] are risk factors for respiratory problems in people with DS. Sleep disorders have in turn been associated with cognitive deterioration due to deficient rest and lack of oxygen,[13] as well as with higher burden and deposition rates of Aβ[14,15] and tau proteins[16,17] due to a poor protein clearance, further contributing to an earlier onset of AD in people with DS.[18]

Diagnosis of AD and OSA is challenging in the general population, but particularly in individuals with DS. Research on biomarkers based on the analysis of protein concentration, such as Aβ, tau, and neurofilament light (NFL) in the cerebrospinal Fluid (CSF) showed a high diagnostic potential for AD.[19,20] However, CSF extraction is invasive and poorly tolerated by people with DS. Concerning OSA detection, an overnight polysomnography (PSG) study is the established gold standard.[21] However, this procedure can be tedious, cause discomfort, and has been questioned as a diagnostic tool in the DS population.[12] Similar to CSF extraction, performing a PSG in people with DS presents additional challenges, since the presence of craniofacial dysmorphologies, obesity, and restless behavior during testing can alter the PSG output, misdiagnosing the presence and severity of OSA.[21]

In the search for complementary biomarkers that are reliable and noninvasive to diagnose the presence of AD and OSA, facial morphology has emerged as a potential biomarker.[22–27] Recent advancements in imaging, computational, and artificial intelligence technologies have developed faster and more reliable tools for diagnosing genetic, rare, neurodevelopmental, and psychotic

15306860, 2025, 6, Downloaded from https://faseb.onlinelibrary.wiley.com/doi/10.1096/fj.202403009R by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [25/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

disorders.[22,23,26,28] These tools use high-throughput methods based on 2D and 3D image analysis to identify phenotypic biomarkers through comparisons of facial patterns between patients and healthy controls.[22–27] Assuming facial development is closely integrated with brain and upper airway development through common signaling pathways,[29,30] previous research underscored that facial traits may indirectly reflect neurological and respiratory disorders in the general population.

Specifically for AD, previous studies have demonstrated that automated computational methods can noninvasively detect dementia using facial traits, with an accuracy above 80% in 2D frontal images[31] and 66% with 3D facial meshes.[32] Regarding OSA, previous research has been able to detect this respiratory condition using noninvasive facial biomarkers with 2D images based on cephalometric parameters[33–36] as well as with 3D meshes, revealing significant local differences concentrated in the nasal region in patients with OSA.[37] Although these studies showed promising results that reinforce the reliability of using facial biomarkers for detecting AD and OSA, they overlooked significant factors that influence facial shape variation, such as sex, age, and facial size. Moreover, these methods have not been assessed in individuals diagnosed with genetic conditions who are at an elevated risk of developing AD and OSA, such as the DS population.

To fill these gaps and to further improve the use of facial biomarkers for early detection and prognosis of neurodevelopmental and respiratory disorders, we conducted the first comprehensive quantitative 3D analysis of facial shape variation, including euploid controls (EU) and individuals diagnosed with DS, AD, and OSA. First, we assessed the effect of age, sex, facial size, and genetic factors on facial shape to understand how these critical factors modulate the facial phenotype. Second, we explored facial adult ontogeny in healthy controls and tested whether this process is altered in DS individuals depending on sex. This is crucial in the investigation of the diagnostic potential of facial biomarkers because the mechanisms that regulate the process of facial development, growth, and aging vary during the ontogenetic trajectory, among individuals, and between sexes.[38–40]

To address these issues, we applied quantitative geometric morphometric analyses,[41–43] which have been previously used to identify disease-and sex-specific facial features associated with genetic and neurodevelopmental disorders. In an earlier study, we reported subtle but significant facial shape differences between healthy controls and patients diagnosed with schizophrenia and bipolar disorder, showing that these facial traits were significantly different between disorders and between male and female patients within each disorder.[28] These results underscore the sensitivity of facial biomarkers for disease diagnosis. In this study, we accounted for the key factors modulating facial shape to assess the potential of facial biomarkers for detecting sex-specific facial traits associated with AD and OSA in the general and the DS populations.

2 | MATERIALS AND METHODS

2.1 | Sample composition

The sample analyzed in this cross-sectional study comprised 347 individuals with and without a diagnosis of AD, including 216 euploid individuals (EU) (72 males, 144 females) and 131 individuals with Down syndrome (DS) (80 males, 51 females) (Table 1). For all individuals, a head magnetic resonance image (MRI) was available, along with information regarding dementia stage and diagnosis of Alzheimer's disease (AD). Data on obstructive sleep apnea (OSA) was also available for a subsample of individuals (Table 1). See Figure S1 for more detailed information on the sample composition. We conducted a power analysis to determine the minimum number of subjects required to assess facial shape differences between EU and DS populations. The results indicated that the minimum required sample size, with a power of 0.8, was 20 individuals for EU controls and 10 for individuals with DS. Our sample overly exceeds these minimum requirements, even when stratified by diagnosis and sex (Figure S1).

All individuals with DS were recruited from the population-based Down-Alzheimer Barcelona Neuroimaging Initiative (DABNI) cohort. Cognitively unimpaired EU participants were recruited from the Sant Pau Initiative on Neurodegeneration (SPIN) cohort, while EU participants with AD were selected from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). All participants were of European origin, with ages ranging from 18 to 62 years (individuals with DS) and from 18 to 90 years (EU controls). The study was approved by the Sant Pau Ethics Committee following the standards for medical research in humans recommended by the Declaration of Helsinki. All participants or their legally authorized representatives gave written informed consent before enrollment. To use the ADNI database, we followed an application protocol that, after evaluation, provided us access to the clinical and imaging data of ADNI participants, who signed consent forms approved by the Resource Allocation Review Committee (RARC).

15306860, 2025, 6, Downloaded from https://faseb.onlinelibrary.wiley.com/doi/10.1096/fj.202403098R by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [25/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 Sample composition.

	AD				
	Sample size				
	No Alzheimer	Alzheimer	Total	Age range (median)	Median ratio Aβ42/40
EU	169 (50/119)	47 (22/25)	216	18–90 (57.07)	0.04–0.121 (0.104)
DS	82 (49/33)	49 (31/18)	131	18–62 (42.51)	0.026–0.116 (0.069)
Total	251	96	347		

	OSA					
	Sample size					
	No	Mild	Severe	Total	Age range (median)	Median AHI
EU	20 (2/18)	5 (2/3)	2 (0/2)	27	18–90 (53.21)	0–38.1 (3)
DS	22 (11/11)	30 (15/15)	30 (26/4)	82	18–62 (40.51)	0–120 (20)
Total	42	35	32	109		

Note: EU and DS groups are divided by AD dementia stages and OSA level. The number of males and females is indicated in brackets as (males/females).

2.2 | AD diagnosis

In the DS cohort, AD diagnosis was established after neurological and neuropsychological examinations with a semi-structured health questionnaire (Cambridge Examination for Mental Disorders of Older People with Down Syndrome and others with intellectual disabilities [CAMDEX-DS]),[44] and a neuropsychological battery including the Cambridge Cognition Examination (CAMCOG). Both tests were adapted for intellectual disabilities, as in Fortea et al.[19] In the ADNI EU cohort, AD diagnosis was obtained following the National Institute on Aging –Alzheimer's Association guidelines for the neuropathologic and neuropsychological assessments of AD.[45]

In both cohorts, the neuropsychological evaluations were accompanied by molecular data, including the concentration ratio of $A\beta_{1\text{-}42}/A\beta_{1\text{-}40}$ proteins obtained from CSF samples. For each individual, the degree of AD dementia was classified as "No Alzheimer" when no signs of AD dementia were detected or as "Alzheimer disease" when individuals showed prodromal or advanced signs of AD dementia (Table 1), following the criteria in Fortea et al.[19]

2.3 | OSA diagnosis

To score the presence and severity of OSA, we used the apnea–hypopnea index (AHI) obtained from nocturnal polysomnography tests performed in sound-attenuated and temperature regulated rooms in the sleep laboratory at the Sant Pau sleep unit following the protocols described in Giménez et al.[30] AHI is defined as the sum of all apneas (airflow reduction greater than 90% for more than 10 seconds) and hypopneas (airflow reduction greater than 30% for at least 10 seconds with an oxygen saturation decrease of ~3% or a cortical awakening) per hour of sleep.[7] Individuals were classified into three categories according to the severity of sleep disruption: (a) no OSA (AHI: <5), (b) mild OSA (AHI: 5–30) and (c) severe OSA (AHI: >30) (Table 1). To streamline the analysis and increase sample size for each group, the "mild OSA category" contained individuals with mild OSA (AHI 5–15) and moderate OSA (AHI 15–30).

2.4 | Facial 3D models and landmarking

3D facial models were generated from MR images (Figure S2) by adjusting the gray-scale threshold that optimized skin segmentation using Amira 5.2.1 (v6.3, Visualization Sciences Group, FEI). The facial models showed similar reconstruction quality and were derived from T1-weighted images obtained from Hospital del Mar (Barcelona, Spain) or Hospital Clinic (Barcelona, Spain), and the ADNI database. Since all individuals were scanned using a 3 T scanner with similar scanning protocols (Table S1), potential experimental bias due to different scan origins was considered negligible.

To capture facial shape, we used an automatic method to record the 3D coordinates of 21 anatomical landmarks on the 3D facial models generated from MR images (Figure 1 and Table S2). The automatic landmarking method is included in *BioFace3D*, a pipeline for the automatic computation of potential facial

15306860, 2025, 6, Downloaded from https://faseb.onlinelibrary.wiley.com/doi/10.1096/fj.202403098R by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [25/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

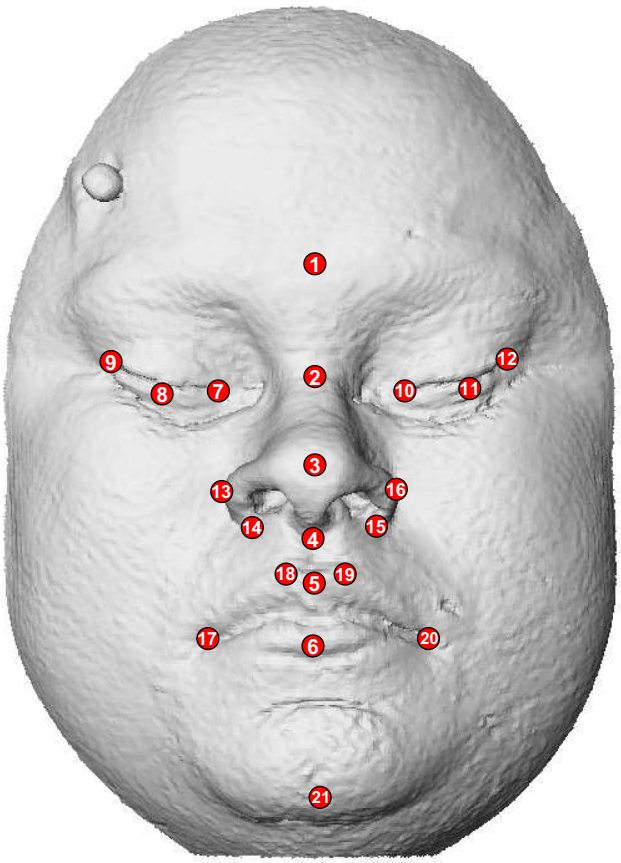

FIGURE 1 Anatomical position of the facial landmarks used for geometric morphometric and multivariate statistical analyses. For landmark definitions see Table S2. This facial model is a morphing which does not belong to a real individual and is based on 3D landmark coordinates.

biomarkers from 3D head models reconstructed from MRI scans.[46] *Bioface3D*[46] applies a domain adaptation and architecture reuse approach[47] on Multi-View Consensus Convolutional Neural Networks (MV-CNN), a deep learning-based state-of-the-art method devised for automatic 3D facial landmarking.[48] This method is based on creating 2D views of the 3D anatomical facial models from different viewpoints. Then, the hourglass model is used to predict candidate landmarks on 2D heat maps, and the final position is determined by ray-projecting onto the surface of the 3D mesh using least squares combined with random sample consensus.[48] The model was trained with datasets of 21 landmarks manually collected on 538 facial meshes, including healthy controls and subjects with genetic and neurodevelopmental disorders.[46] This automatic facial landmarking method has an average landmarking error of 1.78 mm. This accuracy level is lower than 2 mm, which is the intra-observer measurement error commonly accepted for craniometric measurements.[49]

2.5 | Geometric Morphometrics and multivariate statistical analyses

To quantify facial shape variation, we used geometric morphometric (GM) analyses. First, we performed a generalized procrustes analysis (GPA)[50] to superimpose all the configurations of landmarks into a common morphospace by scaling, rotating, and translating the landmark coordinates.[51] As the resulting Procrustes coordinates still contained an allometric effect, we computed the centroid size (CS) of the facial configurations of landmarks and used this variable[51] to account for the effect of facial size on shape in all subsequent analyses.

To quantify the amount of facial shape variation attributable to sex, age, facial size, and genetic diagnosis, we implemented Procrustes ANOVA and MANOVA tests[52,53] using the function ProcD.lm in the Geomorph R package with 1,000 permutations and enforcing the randomized residuals in a permutation procedure (RRPP).[53,54] When sex, age, or facial size were significant factors influencing facial shape variation, subsequent analyses were adjusted using the residuals derived from the Procrustes MANOVA. Analyses were performed combined and separately for EU and DS samples. When sex differences were detected, further analyses were also performed separately for males and females.

Further Procrustes ANOVA tests were performed to assess whether biomarkers for AD (concentration ratio $A\beta_{1\text{-}42}/A\beta_{1\text{-}40}$) and OSA (AHI) explained a significant amount of facial shape variation. Additionally, multivariate linear regressions were performed to further test, statistically and visually, the correlation between facial shape, age, concentration ratio $A\beta_{1\text{-}42}/A\beta_{1\text{-}40}$, and AHI.

To assess statistically significant facial shape differences between diagnostic groups, we computed the Procrustes distances between the average shapes of each group and performed permutation tests with 1,000 iterations.[50] Facial comparisons were performed between the following diagnostic groups: (a) DS/EU, (b) Alzheimer/No Alzheimer in EU and DS groups, and (c) no OSA/mild OSA/severe OSA in the DS group. To visualize the facial shape differences between groups, we conducted canonical variate analysis (CVA), a statistical discriminant method that maximizes the separation among predefined groups.[55,56] Facial morphings and wireframes associated with the negative and positive extremes of the first two CVs were created with IDAV Landmark Editor version and MorphoJ version 1.07a b[57] respectively. Moreover, we created heatmaps using Amira 5.2.1 (v6.3, Visualization Sciences Group, FEI) to identify the facial regions where differences between diagnostic groups were mainly located.

15306860, 2025, 6, Downloaded from https://faseb.onlinelibrary.wiley.com/doi/10.1096/fj.202403098R by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [25/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

All GM and multivariate statistical analyses were executed using *MorphoJ version 1.07a b*[57] and *Geomorph R-Package version 4.0.5*.[54]

3 | RESULTS

3.1 | Sex, age, and facial size influence facial shape

Procrustes ANOVA analyses revealed that sex ($Z = 4.0879$, $p = .001$), age ($Z = 7.4097$, $p = .001$), and facial size ($Z = 7.1928$, $p = .001$) were significant factors influencing facial shape in our sample (Table S3). After adjusting for these factors, the Procrustes MANOVA test also indicated a significant influence of trisomy on facial shape variation ($Z = 6.9455$, $p = .001$) (Table S4). Comparative shape analysis based on Procrustes distances confirmed significant facial differences between EU controls and individuals with DS ($p < .0001$), whereas the CVA showed that trisomic individuals exhibited shorter and wider faces, midfacial retrusion with flattened nasal bridges, and oblique palpebral fissures (Figure 2A,B), which are the facial features typically associated with DS.

3.2 | Sex-dependent effect of trisomy 21 on facial shape

To determine whether trisomy 21 differently affected males and females, we performed a Procrustes MANOVA adjusting only for age and facial size. The results showed that the interaction between genetic diagnosis and sex was statistically significant ($Z = 3.7046$, $p = .002$) (Table S5), suggesting a sex-dependent effect of trisomy 21 on facial shape.

To further explore these sex-specific effects, we performed independent analyses for males and females. Procrustes distances between diagnostic groups confirmed significant facial shape differences between individuals with DS and EU controls in both males ($p < .0001$) and females ($p < .0001$). The CVAs showed that individuals diagnosed with DS from both sexes showed the typical DS facial features (Figure 2C–F). However, the heatmaps and wireframes representing facial shape differences between EU controls and DS individuals showed that DS facial traits were more pronounced in females (Figure 2E,F).

3.3 | Altered facial ontogenetic trajectories in females with Down syndrome

After evaluating the effect of sex on facial shape, we compared how facial shape changes over adulthood in EU and DS groups. A multivariate linear regression analysis revealed a significant relationship of facial shape with age (%Predicted = 7.11%, $P \leq .0001$) and showed the effects of aging in both groups. Since the slope of the regressions was not parallel, the analysis suggested differences in the ontogenetic trajectories of EU controls and individuals diagnosed with DS (Figure 3A). A Procrustes MANOVA confirmed a significant interaction between genetic diagnosis and age after adjusting for sex and facial size ($Z = 1.5826$, $p = .005$) (Table S6A), demonstrating that facial adult ontogeny is altered in DS. Individuals with DS presented a steeper slope of the ontogenetic trajectory, showing more pronounced facial shape changes during aging in comparison to EU controls (Figure 3A).

To further explore sex-specific facial shape changes in the ontogenetic trajectories between EU controls and individuals with DS, we repeated the multivariate regressions and Procrustes MANOVA tests separately for males and females, adjusting only for facial size. In the male sample, the multivariate regression results showed a significant correlation between facial shape and age among EU and DS individuals (%Predicted = 8.79%, $p \leq .0001$) (Figure 3B). However, the Procrustes MANOVA showed that the interaction between diagnosis and age was not significant ($Z = -0.3126$, $p = .605$) (Table S6B, Figure 3B), indicating no differences in the ontogenetic trajectories between the EU and DS groups in males. On the contrary, in the female sample, we detected both a statistical significance in the correlation between facial shape and age (%Predicted = 6.98%, $p \leq .0001$) (Figure 3C) and in the interaction between diagnosis and age ($Z = 1.8775$, $p = .033$) (Table S6C, Figure 3C). These results indicated that the altered ontogenetic trajectory in the DS population (Figure 3A) was only due to female individuals (Figure 3C), who showed more pronounced facial aging effects.

3.4 | Significant correlations of facial shape with AD and OSA

Considering the previous results, we accounted for the effect of sex, age, and facial size in the subsequent analyses assessing the potential of facial shape as a diagnostic biomarker for AD and OSA in the DS and EU populations.

Regarding AD, we first investigated the association between facial shape variation and molecular biomarkers of dementia. A multivariate regression analysis indicated that the concentration ratio of $A\beta_{1\text{-}42}/A\beta_{1\text{-}40}$ in CSF was significantly correlated with facial shape after adjusting for sex, age, and facial size when EU controls and individuals

15306860, 2025, 6, Downloaded from https://faseb.onlinelibrary.wiley.com/doi/10.1096/fj.202403098R by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [25/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

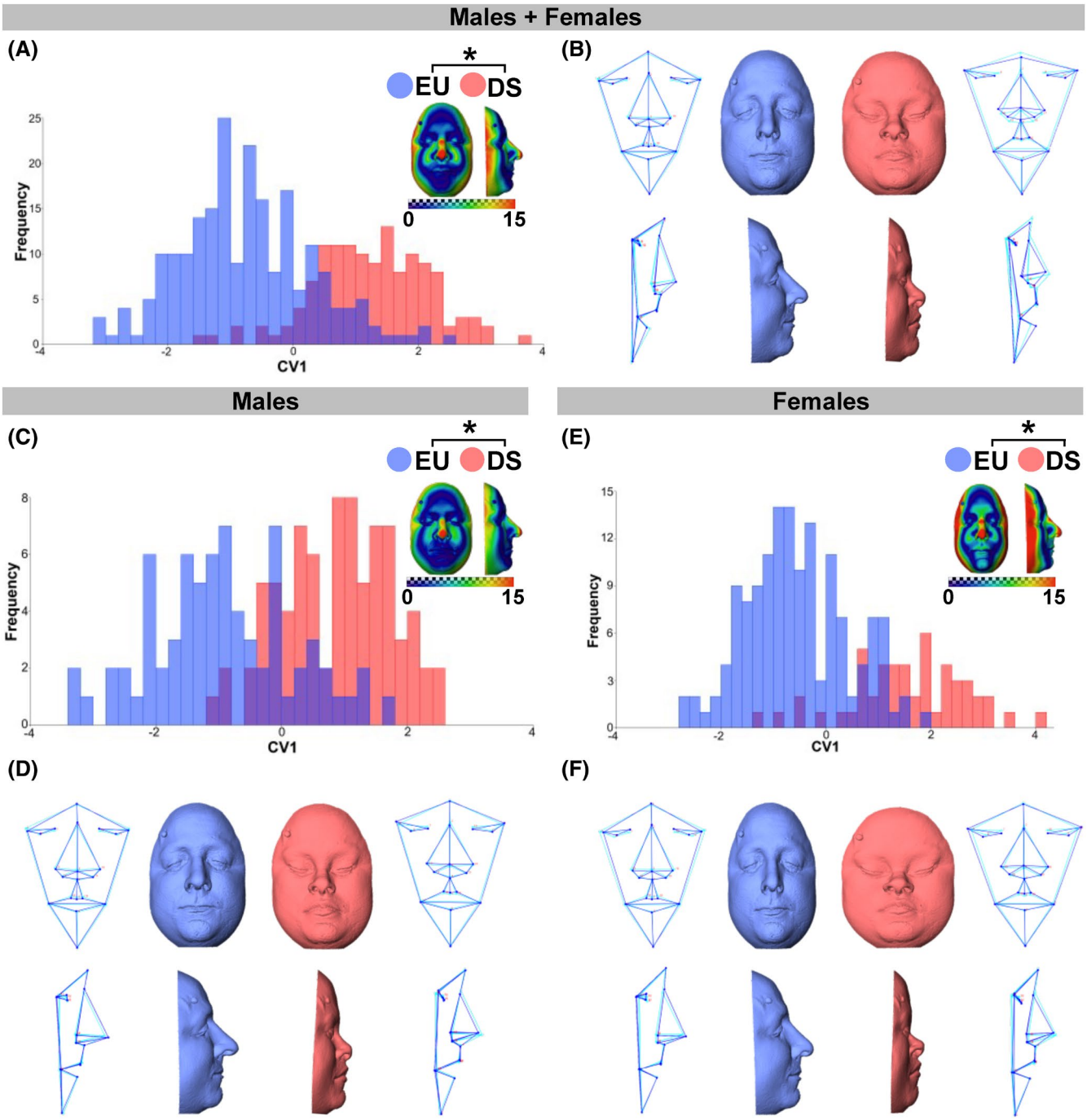

FIGURE 2 Results of the canonical variate analyses (CVA) between EU controls and individuals with DS. Analyses were performed for three groups: Males and females (A, B), only males (C, D), and only females (E, F). Panels A, C, and E display Canonical Variate1 (CV1), accounting for 100% of facial shape variation. Panels B, D, and F show facial morphings and wireframes associated with the negative extreme of CV1 (representing frontal and lateral views of EU individuals in blue), and the positive extreme of CV1 (representing individuals diagnosed with DS in red). Facial morphings are magnified by a factor of 2. CVA results are accompanied by heatmaps illustrating facial shape differences between the morphings at the positive and negative extremes of CV1. Red areas indicate anatomical regions with largest facial shape differences between EU controls and individuals with DS, whereas blue areas indicate smallest facial shape differences. *Statistically significant difference between groups ($p < .0001$). The faces in the figure are morphings which do not belong to a real individual and are based on 3D landmark coordinates.

with DS were pooled together (%Predicted = 4.4%, $p = .001$) (Table S7, Figure 4). However, when the association was evaluated separately for each group, the multivariate regressions did not show significant correlations either for EU controls (%Predicted = 1.8%, $p = .369$) or for individuals with DS (%Predicted = 2.1%, $p = .689$).

15306860, 2025, 6, Downloaded from https://faseb.onlinelibrary.wiley.com/doi/10.1096/fj.202403098R by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [25/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

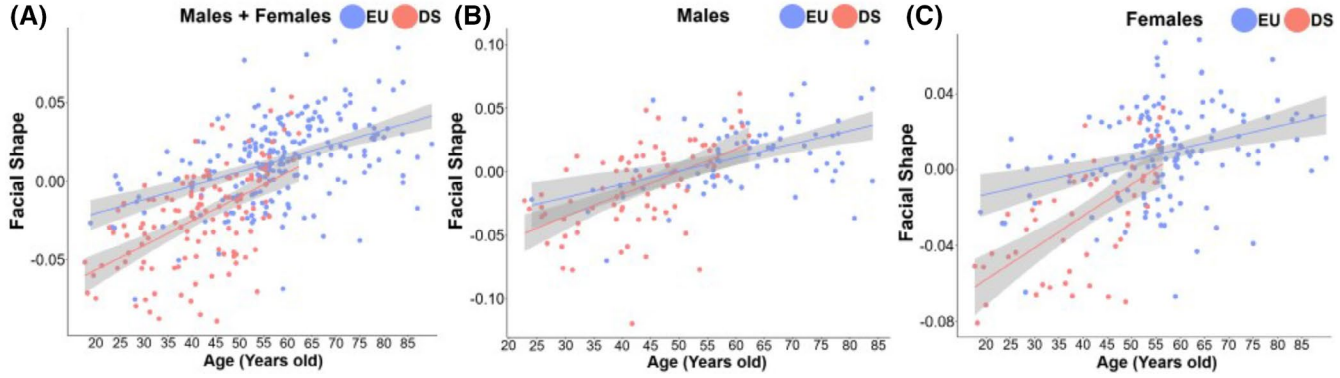

FIGURE 3 Ontogenetic trajectories of EU and DS populations assessed by multivariate regression of facial shape on age. (A) Male + Female, (B) Male and (C) Female groups.

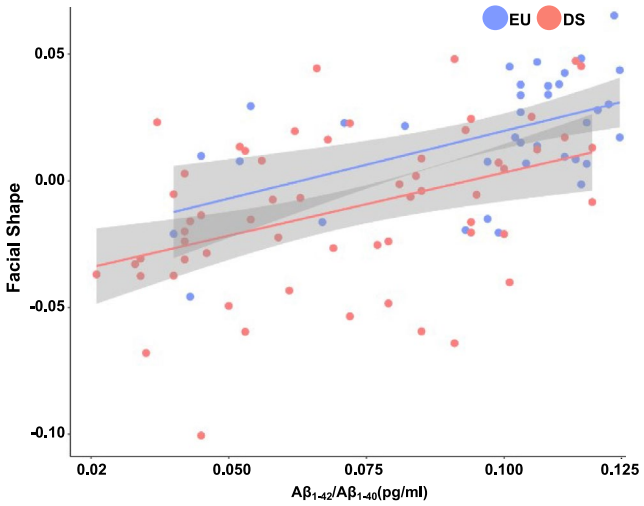

FIGURE 4 Multivariate regression of facial shape on concentration ratio $A\beta_{1\text{-}42}/A\beta_{1\text{-}40}$, an AD diagnostic biomarker.

Second, we explored the facial morphological variation associated with the diagnosis of AD in each group, and separately for males and females. In the DS population, facial shape differences between individuals with and without an AD diagnosis were significant before adjusting for age, sex, and facial size ($Z = 2.3385$, $p = .008$). However, once adjusted for those variables, statistical significance was not achieved ($Z = -0.30077$, $p = .611$) (Figure 5A). Moreover, when we performed the analyses separately for each sex, we did not detect any significant facial shape differences between individuals with and without AD in males ($Z = 0.21119$, $p = .411$) (Figure 5C), nor in females ($Z = 0.072242$, $p = .468$) (Figure 5E).

In the EU population, significant shape differences were detected between individuals with and without an AD diagnosis before ($Z = 4.4324$, $p = .001$) and after ($Z = 3.5777$, $p = .001$) (Figure 6A) adjusting for age, sex, and facial size. We found that the main facial morphological differences between EU controls with and without AD were independent of age, with facial changes located in the facial midline, specifically on the forehead and chin (Figure 6A,B). Moreover, we identified significant facial differences between individuals with and without AD dementia in both males ($Z = 2.2095$, $p = .015$) (Figure 6C) and females ($Z = 2.4276$, $p = .009$) (Figure 6E). In males, the main facial differences were detected in the facial midline (Figure 6D), whereas in females, the differences were localized on the forehead and the facial width (Figure 6F).

Regarding OSA, we explored the association between facial shape variation and a clinical score of sleep apnea in a subsample of individuals. The multivariate regression of facial shape variation on AHI after adjusting for age, sex, and facial size demonstrated a significant correlation when the EU and DS populations were pooled together (%Predicted = 3.4%, $Z = 1.8139$, $p = .039$) (Figure 7). When the association was tested separately for each group, the multivariate regressions indicated no significant correlation for EU controls (%Predicted = 2.1%, $p = .902$), but a significant correlation for individuals with DS (%Predicted = 2.9%, $p = .009$). A Procrustes ANOVA confirmed that AHI accounted for a small but significant percentage of facial shape variation in the DS group ($Z = 2.1487$, $p = .017$) (Table S8).

Finally, when we assessed whether there were significant facial shape differences between individuals diagnosed with DS and different levels of OSA severity, the Procrustes distances between groups revealed that individuals with DS and severe OSA presented a significantly different facial morphology than individuals with DS and no OSA ($p = .005$). The CVA showed that, in comparison to individuals with no signs of OSA, individuals with DS and severe OSA were characterized by narrower faces and

15306860, 2025, 6, Downloaded from https://faseb.onlinelibrary.wiley.com/doi/10.1096/fj.202403069R by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [25/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

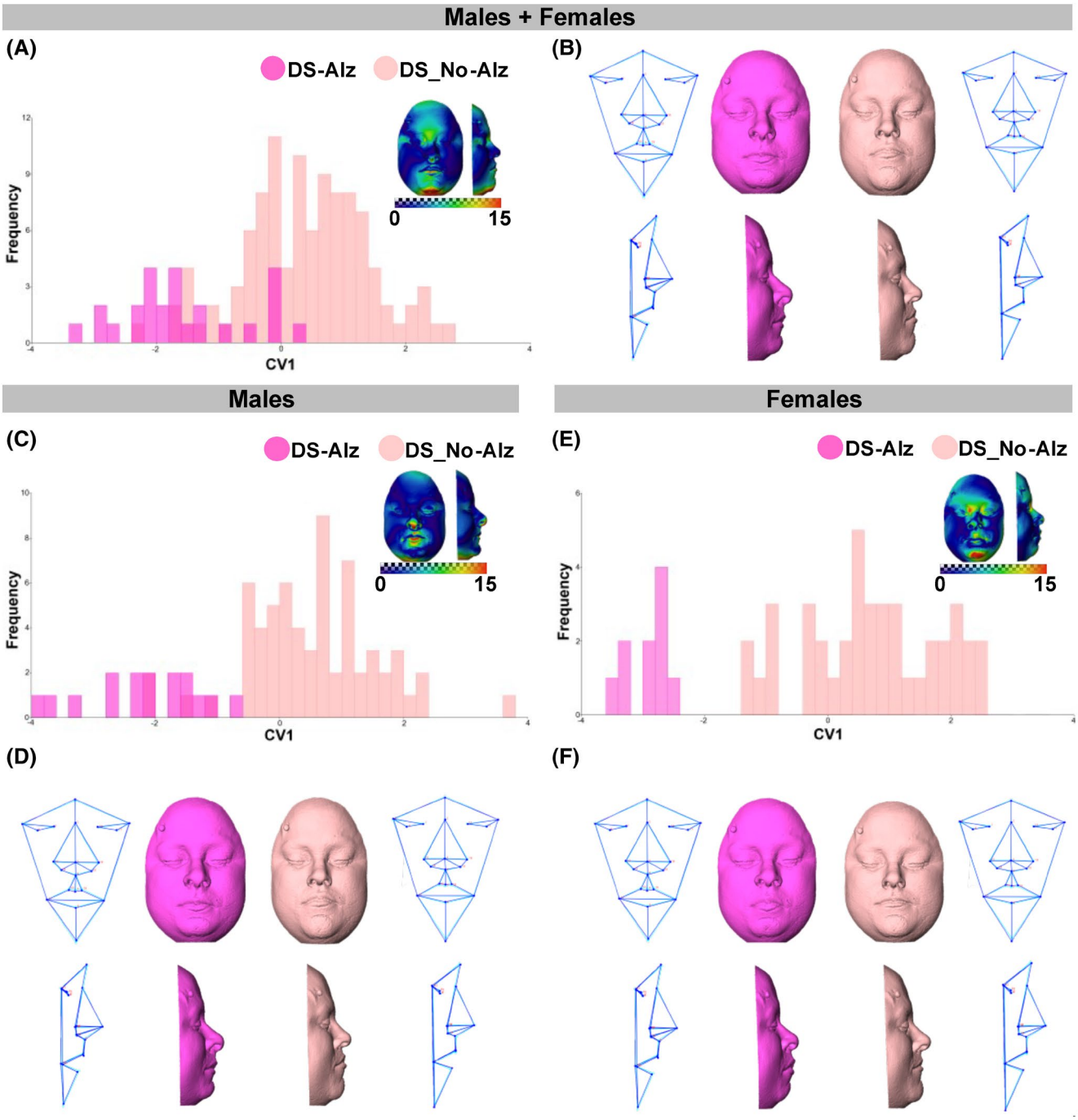

FIGURE 5 Results of the canonical variate analyses (CVA) between DS individuals with and without AD dementia. Panels A, C, and E display Canonical Variate1 (CV1), accounting for 100% of facial shape variation. Panels B, D, and F show facial morphings and wireframes associated with the negative extreme of CV1 (representing frontal and lateral views of DS individuals with AD dementia in dark pink), and the positive extreme of CV1 (representing DS individuals without AD dementia in light pink). Red areas of the heatmaps indicate anatomical regions with largest facial shape differences between DS individuals with and without AD dementia, whereas blue areas indicate smallest facial shape differences. For more details see legend in Figure 2.

subtle mandibular retraction (Figure 8A,B). Individuals with mild OSA fell between the individuals with no signs of OSA and individuals with severe OSA, showing a continuum spectrum of facial shape variation associated with OSA and suggesting a relationship between facial morphology and OSA worsening.

4 | DISCUSSION

Our study is the first to perform a comprehensive and quantitative 3D analysis of facial shape variation in Down syndrome (DS) and euploid control (EU) populations to assess the diagnostic potential of facial biomarkers for Alzheimer's

15306860, 2025, 6, Downloaded from https://faseb.onlinelibrary.wiley.com/doi/10.1096/fj.202403009R by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [25/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

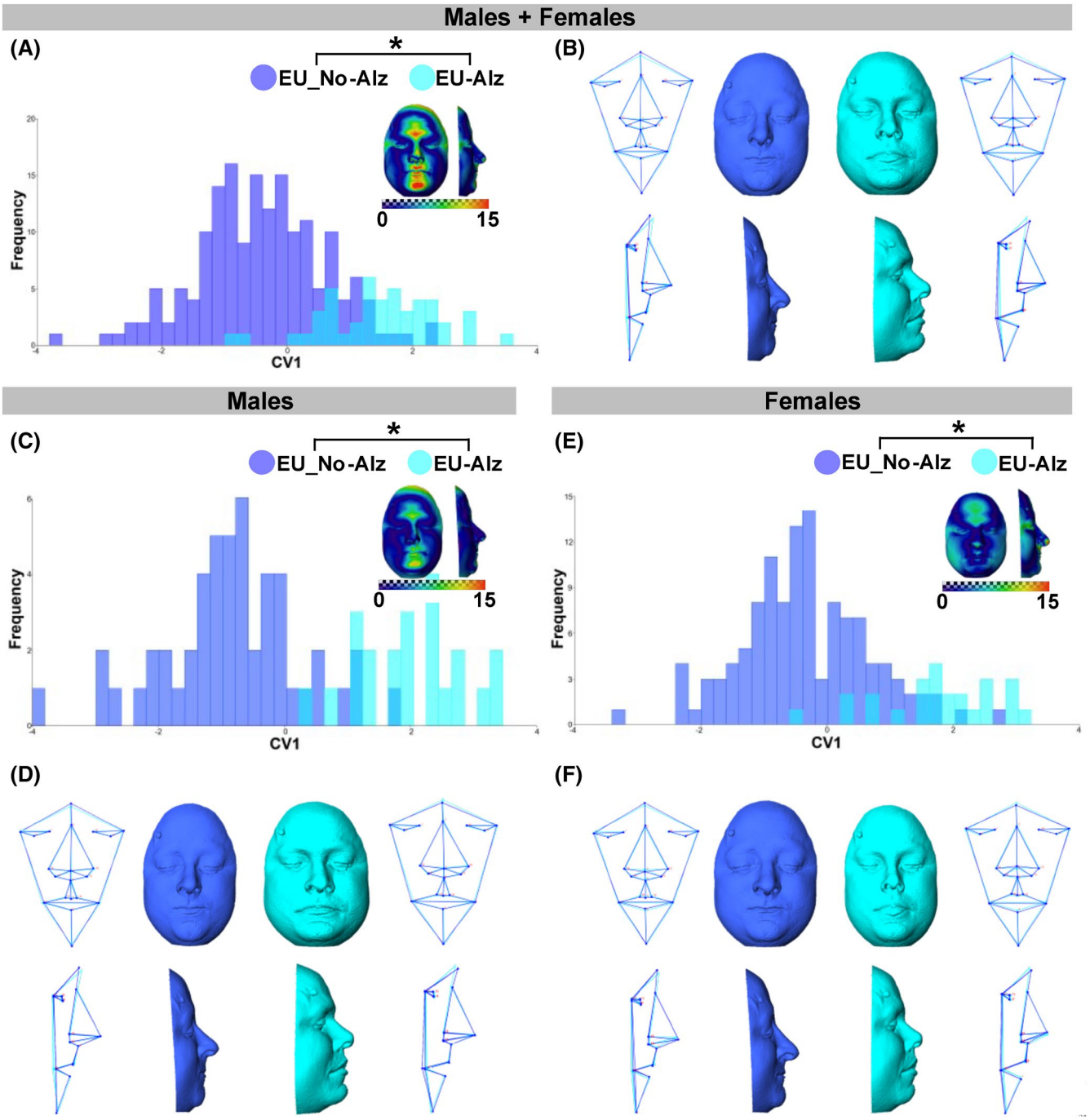

FIGURE 6 Results of the canonical variate analyses (CVA) between EU controls with and without AD dementia. Panels A, C, and E display Canonical Variate1 (CV1), accounting for 100% of facial shape variation. Panels B, D, and F show facial morphings and wireframes associated with the negative extreme of CV1 (representing frontal and lateral views of EU controls without AD dementia in dark blue), and the positive extreme of CV1 (representing EU controls with AD dementia in light blue). Red areas of the heatmaps indicate anatomical regions with largest facial shape differences between EU controls with and without AD dementia, whereas blue areas indicate smallest facial shape differences. For more details see legend in Figure 2.

disease (AD) and obstructive sleep apnea (OSA). Our results demonstrated that age, sex, and facial size are significant factors in facial shape variation, and that these factors should be accounted for to optimize the diagnostic potential and accuracy of facial biomarkers. Furthermore, we demonstrated that facial shape showed diagnostic and sex-specific patterns. First, we identified more severe facial traits in females with DS. Second, we detected significant facial patterns associated with AD in EU and OSA in DS populations. Overall, our results support the use of facial shape as an indirect biomarker for neurodegenerative and respiratory disorders in future studies and clinical applications.

15306860, 2025, 6, Downloaded from https://faseb.onlinelibrary.wiley.com/doi/10.1096/fj.202403098R by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [25/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

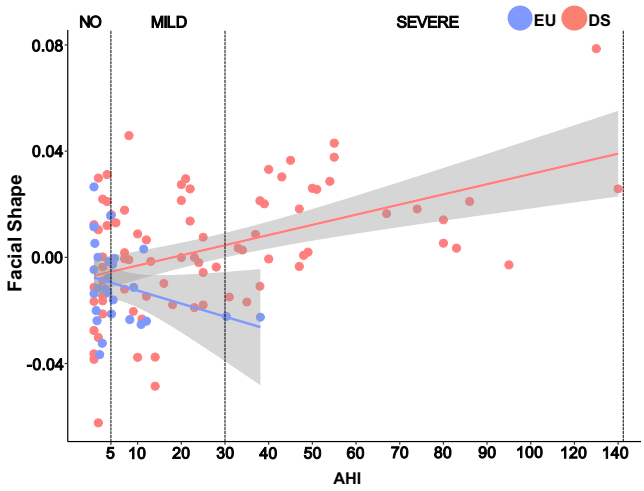

FIGURE 7 Multivariate regression of facial shape on AHI, and OSA diagnostic biomarker.

Our findings showed that after adjusting for the significant effect of age, sex, and facial size, individuals diagnosed with DS displayed a significantly different facial morphology as compared with EU controls. The facial traits associated with DS in our study coincided with the typical DS facial features described in previous studies,[10,11,58] including shorter and wider faces, midfacial retrusion, flattened nasal bridges, and oblique palpebral fissures. However, our study provided original evidence that trisomy 21 differentially affects the facial morphology of males and females diagnosed with DS, with females showing more pronounced facial shape differences. Other studies have reported sex-related differences in individuals with DS, such as an earlier decline in bone mineral density (BMD) in males[59,60] and a slower rate of BMD acquisition in females.[61] Additionally, females with DS tend to have higher fat levels and lower lean mass, which carry significant health implications.[62] Given that facial anatomy is composed of both bone and soft tissue (fat and lean mass), these previous findings[59–62] suggest that the reduced BMD and lean mass in females could contribute to the more pronounced DS-associated facial features observed in this population.

Our results also evidenced facial shape changes over adult ontogeny in both the EU and DS populations. These findings are consistent with previous studies indicating that age had a significant effect on facial shape in the general[63] and the DS populations.[64,65] Specifically, these studies showed that individuals with DS exhibit specific facial patterns that appear early in life but can vary in adolescence or adulthood, becoming either subtler or more pronounced.[64,65] Our results showed that adults with DS retained the characteristic facial features associated with trisomy 21, but only females showed a significant deviation of facial adult ontogeny compared with EU controls, displaying more pronounced facial aging effects. Our results thus support previous research showing that even though there is an early genetic contribution establishing the typical facial morphology of diverse syndromes, postnatal ontogeny plays a key role in facial morphological variation throughout life.[40] Moreover, this result aligns with our previous findings, which indicate a more severe phenotype in females with DS, particularly in early adulthood. We detected significant differences in the slopes of the ontogenetic trajectories in females, showing more severe DS phenotypes during adolescence that vary with aging.

Regarding the association between facial morphology and AD, our study demonstrated that when EU controls and individuals with DS were pooled together, facial shape significantly correlated with Aβ1-42 concentration in CSF, which has been proposed as an accurate diagnostic biomarker for AD.[19] However, this association was not observed when EU controls and individuals with DS were analyzed separately, possibly due to the limited sample size. Furthermore, we detected significant facial shape differences between individuals with and without AD in the EU population that were sex-specific and independent of age. These results highlight the need for sex-differential approaches for AD detection using facial biomarkers, and suggest that facial morphology could be used as a noninvasive and complementary biomarker for detecting AD. This supports the findings in Umeda-Kameyama et al.[31] and Zheng et al.,[32] who demonstrated that 2D[31] and 3D[32] facial morphology accurately assist in the noninvasive detection of AD dementia.

However, the significant facial shape differences between individuals with and without AD that were detected in the DS population were not consistent after adjusting for age, sex, and facial size. With the available data, we cannot determine whether this result is due to experimental bias from limited sample sizes in the DS population or due to biological mechanisms associated with DS and AD. It is likely that the facial dysmorphologies caused by trisomy 21 are so pronounced that they may mask the facial shape changes associated with AD. Further studies are needed to confirm this hypothesis.

Concerning the association between facial shape and OSA, our study indicated a significant correlation between facial morphology and AHI, the most commonly used biomarker for diagnosing OSA.[7] Furthermore, when analyzing the facial morphological variation associated with OSA, our results demonstrated that individuals diagnosed with DS and severe OSA displayed a significantly different facial shape compared with individuals with DS and no signs of OSA. The facial morphology of individuals with DS affected by severe OSA was characterized by narrower faces and a slight mandibular retraction, which can lead to restricted upper airways and a minor backward displacement of

15306860, 2025, 6, Downloaded from https://faseb.onlinelibrary.wiley.com/doi/10.1096/fj.202403009R by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [25/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

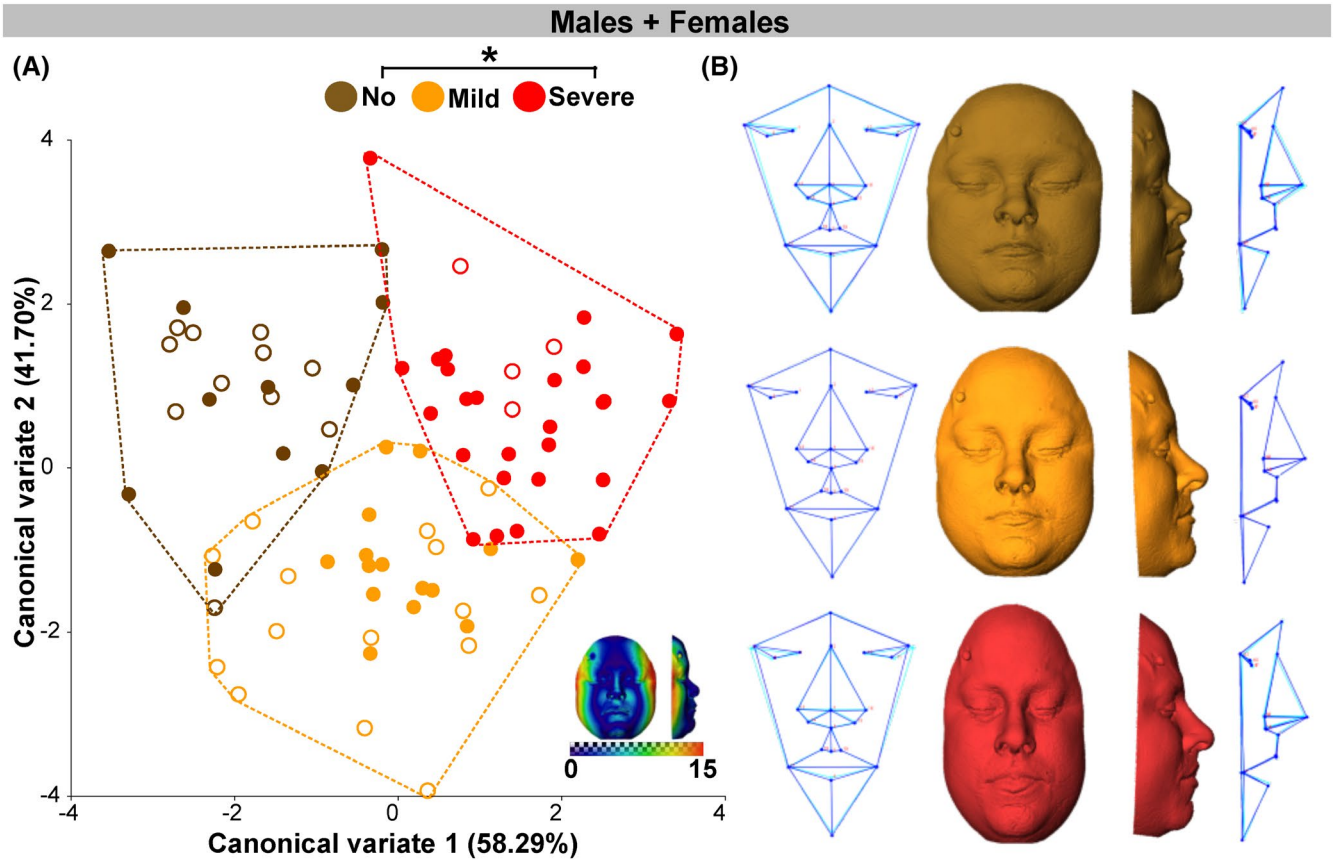

FIGURE 8 Results of the CVA assessing facial differences between individuals with DS and different OSA levels. Panel A displays CVA results with heatmaps illustrating facial shape differences between individuals with DS at the positive extreme (severe OSA) and negative extreme (no OSA) of CV1. Red areas indicate anatomical regions with the largest facial shape differences, while blue areas indicate the smallest. Panel B presents facial morphings and wireframes representing individuals with DS at different OSA levels along CV1. *Statistically significant difference between groups ($p < .0001$). The faces in the figure are morphings which do not belong to a real individual and are based on 3D landmark coordinates.

the oropharynx that may explain the higher prevalence of OSA in the DS population. In contrast to previous studies that found no statistically significant differences associated with OSA using 3D photogrammetric measurement[66] or detected only 2D local significant facial differences in the EU population or using linear measurements based on cephalometric parameters,[33–36] our study showed significant differences in the DS population. Therefore, our results demonstrate that 3D facial anatomy is a potential noninvasive biomarker for detecting OSA and for preventing further cognitive impairment in the DS population.

4.1 | Future directions and conclusions

Understanding the bidirectional link between AD and OSA is crucial for the DS population. Several studies have demonstrated that sleep disturbances, including OSA, can increase the risk of an earlier AD onset,[67,68] likely due to poor sleep quality, which impairs neuronal repair and reduces the clearance of proteins such as Aβ[14,15] and tau,[16,17] leading to increased deposition rates. Moreover, other studies have shown that sleep-related problems generally increase as AD progresses.[18,67,69] Currently, there is no effective cure or treatment for AD that can mitigate the progress of this neurodegenerative disorder. Moreover, people with DS are at an ultrahigh risk for suffering from AD due to different genetic and morphological factors, and will invariably develop AD in adulthood.[70–72] However, sleep-related disturbances, such as OSA, display modifiable risk factors that can be treated, minimizing derived neurodegenerative comorbidities such as AD and thereby improving life quality.[7,73]

Our results encourage further research using larger samples to determine whether a significant association between facial shape and AD biomarkers, such as Aβ1-42 concentration, can be detected when EU controls and individuals with DS are analyzed separately. Moreover,

15306860, 2025, 6, Downloaded from https://faseb.onlinelibrary.wiley.com/doi/10.1096/fj.202403098R by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [25/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

a larger sample of individuals with DS is required to assess whether facial shape differences can be identified between those with and without AD dementia. Regarding OSA, larger samples could enable the evaluation of facial shape differences by sex and to determine if significant morphological differences can be detected in individuals with mild OSA. Moreover, additional data of EU controls with OSA are needed to extend our facial morphological analyses to populations with sleep-related disorders but no genetic condition.

Our study concludes that 3D facial morphology can serve as a noninvasive diagnostic biomarker for Alzheimer's disease and obstructive sleep apnea. With adequate adjustments for sex, age, and size, and further research, facial biomarkers could be used in the future for early detection and improved clinical management of these disorders in Down syndrome and the general populations.

AUTHOR CONTRIBUTIONS

LME-Q, YH, XS, JF, and NM-A conceived the project. SG and JF recruited participants, performed MR scanning, cognitive evaluations, polysomnography tests, and diagnosed cases with AD and OSA within the DS cohort. MR-A provided us with the magnetic resonance images and the clinical data associated. AH-L, XS, and LME-Q developed and validated the automatic landmarking method. LME-Q, SL, PI, YH, and NM-A analyzed data and interpreted results. LME-Q, SL, and NM-A prepared the manuscript and figures. All authors critically reviewed the manuscript.

ACKNOWLEDGMENTS

The authors would like to thank all study participants, their families, and caregivers from the DABNI and SPIN cohort for their support of, and dedication to, this research. We also would like to acknowledge the Fundació Catalana Síndrome de Down (https://fcsd.org/) for global support. We also acknowledge the Alzheimer's Disease Neuroimaging Initiative (ADNI) for the data shared. Data collection and sharing for this project was funded by the Alzheimer's Disease Neuroimaging Initiative (ADNI) (National Institutes of Health Grant U01 AG024904) and DOD ADNI (Department of Defense award number W81XWH-12-2-0012). ADNI is funded by the National Institute on Aging, the National Institute of Biomedical Imaging and Bioengineering, and through generous contributions from the following: AbbVie, Alzheimer's Association; Alzheimer's Drug Discovery Foundation; Araclon Biotech; BioClinica, Inc.; Biogen; Bristol-Myers Squibb Company; CereSpir, Inc.; Cogstate; Eisai Inc.; Elan Pharmaceuticals, Inc.; Eli Lilly and Company; EuroImmun; F. Hoffmann-La Roche Ltd. and its affiliated company Genentech, Inc.; Fujirebio; GE Healthcare; IXICO Ltd.; Janssen Alzheimer Immunotherapy Research & Development, LLC; Johnson & Johnson Pharmaceutical Research & Development LLC; Lumosity; Lundbeck; Merck & Co., Inc.; Meso Scale Diagnostics, LLC; NeuroRx Research; Neurotrack Technologies; Novartis Pharmaceuticals Corporation; Pfizer Inc.; Piramal Imaging; Servier; Takeda Pharmaceutical Company; and Transition Therapeutics. The Canadian Institutes of Health Research is providing funds to support ADNI clinical sites in Canada. Private sector contributions are facilitated by the Foundation for the National Institutes of Health (www.fnih.org). The grantee organization is the Northern California Institute for Research and Education, and the study is coordinated by the Alzheimer's Therapeutic Research Institute at the University of Southern California. ADNI data are disseminated by the Laboratory for Neuro Imaging at the University of Southern California.

FUNDING INFORMATION

The research in this article was supported by the Fondation Jérôme Lejeune with grant 2020b cycle-Project No. 2001 and a postdoctoral fellowship to SL (No. 55, 2023 edition), as well as Beca predoctoral Fundación Álvaro Entrecanales-Lejeune to LME-Q and Joan Oró grant (2024 FI-300160) from the Recerca i Universitats Departament (DRU) of the Generalitat de Catalunya and the European Social Fund to AH-L. This study was funded by the Fondo de Investigaciones Sanitario, Instituto de Salud Carlos III and co-funded by the European Union (PI14/01126 and PI17/01019 to JF, and PI20/00836 to SG). This work was also supported by the National Institutes of Health (NIH grants 1R01AG056850-01A1; R21AG056974; R01AG061566 and 1RF1AG080769-01) and Jérôme Lejeune Foundation (JLF1801) to JF, and from the Alzheimer's Association and Global Brain Health Institute (GBHI ALZ UK-23-971 107) to SG. The authors would also like to thank the support from the Agència de Gestió d'Ajuts Universitaris i de Recerca (AGAUR) of the Generalitat de Catalunya (2021 SGR00706 and 2021 SGR01396), the Agencia Española de Investigación (PID2020-113609RB-C21/AEI/10.13039/501100011033), and the International Collaboration and Exchange Program (ICEP).

DISCLOSURES

Mateus Rozalem-Aranha Funding: Alzheimer's Association Research Fellowship to promote Diversity (AARFD-21-852492). Mateus Rozalem-Aranha COI: Payed consultancy to VERANEX. Board participation in Masima – Soluções em Imagens Médicas LTDA.

15306860, 2025, 6, Downloaded from https://faseb.onlinelibrary.wiley.com/doi/10.1096/fj.202403098R by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [25/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

DATA AVAILABILITY STATEMENT

Raw image data cannot be made available due to restrictions imposed by the ethics approval. Landmark data supporting the findings of this study are available from the corresponding authors upon reasonable request.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

All participants or their legally authorized representatives coming from Hospital de la Santa Creu i Sant Pau signed an informed consent before enrollment with the protocols approved by the local ethics committee of Hospital de la Santa Creu i Sant Pau following the standards for medical research in humans recommended by the Declaration of Helsinki. To use the ADNI database, we followed an application protocol that, after evaluation, provided us access to the clinical and imaging data of ADNI participants, who signed consent forms approved by the Resource Allocation Review Committee (RARC).

CONSENT FOR PUBLICATION

Alzheimer's Disease Neuroimaging Initiative (ADNI) provided us the consent for publication after following all the data request processes and reviewing the manuscript.

ORCID

L. M. Echeverry-Quiceno https://orcid.org/0000-0002-9799-0252
S. Llambrich https://orcid.org/0000-0001-9980-0208
Á. Heredia-Lidón https://orcid.org/0000-0001-6325-8744
S. Giménez https://orcid.org/0000-0001-8031-2319
M. Rozalem-Aranha https://orcid.org/0000-0001-9594-292X
P. Inampudi https://orcid.org/0009-0003-0206-1317
Y. Heuzé https://orcid.org/0000-0002-0660-9613
X. Sevillano https://orcid.org/0000-0002-6209-3033
J. Fortea https://orcid.org/0000-0002-1340-638X
N. Martínez-Abadías https://orcid.org/0000-0003-3061-2123

REFERENCES

1. Bittles AH, Glasson EJ. Clinical, social, and ethical implications of changing life expectancy in Down syndrome. *Dev Med Child Neurol*. 2004;46(4):282-286. doi:10.1017/s0012162204000441
2. Roper RJ, Reeves RH. Understanding the basis for Down syndrome phenotypes. *PLoS Genet*. 2006;2(3):e50. doi:10.1371/journal.pgen.0020050
3. Glasson EJ, Sullivan SG, Hussain R, et al. The changing survival profile of people with Down's syndrome: implications for genetic counselling. *Clin Genet*. 2002;62(5):390-393. doi:10.1034/j.1399-0004.2002.620506.x
4. Guralnick MJ. Why early intervention works: a systems perspective. *Infants Young Child*. 2011;24(1):6-28. doi:10.1097/IYC.0b013e3182002cfe
5. Glasson EJ, Dye DE, Bittles AH. The triple challenges associated with age-related comorbidities in Down syndrome. *J Intellect Disabil Res*. 2014;58(4):393-398. doi:10.1111/jir.12026
6. Real de Asua D, Quero M, Moldenhauer F, Suarez C. Clinical profile and main comorbidities of Spanish adults with Down syndrome. *Eur J Intern Med*. 2015;26(6):385-391. doi:10.1016/j.ejim.2015.05.003
7. Giménez S, Videla L, Romero S, et al. Prevalence of sleep disorders in adults with Down syndrome: a comparative study of self-reported, actigraphic, and polysomnographic findings. *J Clin Sleep Med*. 2018;14(10):1725-1733. doi:10.5664/jcsm.7382
8. Wiseman FK, Pulford LJ, Barkus C, et al. Trisomy of human chromosome 21 enhances amyloid-β deposition independently of an extra copy of APP. *Brain*. 2019;142(6):e25. doi:10.1093/brain/awz061
9. Trois MS, Capone GT, Lutz JA, et al. Obstructive sleep apnea in adults with Down syndrome. *J Clin Sleep Med*. 2009;5(4):317-323. doi:10.5664/jcsm.27541
10. Starbuck JM, Cole TM 3rd, Reeves RH, Richtsmeier JT. Trisomy 21 and facial developmental instability. *Am J Phys Anthropol*. 2013;151(1):49-57. doi:10.1002/ajpa.22255
11. Starbuck JM, Llambrich S, Gonzàlez R, et al. Green tea extracts containing epigallocatechin-3-gallate modulate facial development in Down syndrome. *Sci Rep*. 2021;11(1):4715. doi:10.1038/s41598-021-83757-1
12. Hurvitz MS, Lesser DJ, Dever G, et al. Findings of routine nocturnal polysomnography in children with Down syndrome: a retrospective cohort study. *Sleep Med*. 2020;76:58-64. doi:10.1016/j.sleep.2020.10.003
13. Gao F, Wei S, Dang L, et al. Sleep disturbance is associated with mild cognitive impairment: a community population-based cross-sectional study. *BMC Public Health*. 2022;22(1):2000. doi:10.1186/s12889-022-14391-3
14. Yun CH, Lee HY, Lee SK, et al. Amyloid burden in obstructive sleep apnea. *J Alzheimers Dis*. 2017;59(1):21-29. doi:10.3233/JAD-161047
15. Sharma RA, Varga AW, Bubu OM, et al. Obstructive sleep apnea severity affects amyloid burden in cognitively normal elderly. a longitudinal study. *Am J Respir Crit Care Med*. 2018;197(7):933-943. doi:10.1164/rccm.201704-0704OC
16. Carvalho DZ, St Louis EK, Schwarz CG, et al. Witnessed apneas are associated with elevated tau-PET levels in cognitively unimpaired elderly. *Neurology*. 2020;94(17):e1793-e1802. doi:10.1212/WNL.0000000000009315
17. Bubu OM, Pirraglia E, Andrade AG, et al. Obstructive sleep apnea and longitudinal Alzheimer's disease biomarker changes. *Sleep*. 2019;42(6):zsz048. doi:10.1093/sleep/zsz048
18. Emamian F, Khazaie H, Tahmasian M, et al. The association between obstructive sleep apnea and Alzheimer's disease: a meta-analysis perspective. *Front Aging Neurosci*. 2016;8:78. doi:10.3389/fnagi.2016.00078
19. Fortea J, Carmona-Iragui M, Benejam B, et al. Plasma and CSF biomarkers for the diagnosis of Alzheimer's disease in adults with Down syndrome: a cross-sectional study. *Lancet Neurol*. 2018;17(10):860-869. doi:10.1016/S1474-4422(18)30285-0
20. Fortea J, Zaman SH, Hartley S, et al. Alzheimer's disease associated with Down syndrome: a genetic form of

15306860, 2025, 6, Downloaded from https://faseb.onlinelibrary.wiley.com/doi/10.1096/fj.202403098R by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [25/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

dementia. *Lancet Neurol*. 2021;20(11):930-942. doi:10.1016/S1474-4422(21)00245-3
21. Sutherland K, Almeida FR, de Chazal P, Cistulli PA. Prediction in obstructive sleep apnoea: diagnosis, comorbidity risk, and treatment outcomes. *Expert Rev Respir Med*. 2018;12(4):293-307. doi:10.1080/17476348.2018.1439743
22. Echeverry-Quiceno LM, Candelo E, Gómez E, et al. Population-specific facial traits and diagnosis accuracy of genetic and rare diseases in an admixed Colombian population. *Sci Rep*. 2023;13:6869. doi:10.1038/s41598-023-33374-x
23. Hallgrímsson B, Aponte JD, Katz DC, et al. Automated syndrome diagnosis by three-dimensional facial imaging. *Genet Med*. 2020;22(10):1682-1693. doi:10.1038/s41436-020-0845-y
24. Gurovich Y, Hanani Y, Bar O, et al. Identifying facial phenotypes of genetic disorders using deep learning. *Nat Med*. 2019;25(1):60-64. doi:10.1038/s41591-018-0279-0
25. Hsieh TC, Bar-Haim A, Moosa S, et al. GestaltMatcher facilitates rare disease matching using facial phenotype descriptors. *Nat Genet*. 2022;54(3):349-357. doi:10.1038/s41588-021-01010-x
26. Bannister JJ, Crites SR, Aponte JD, et al. Fully automatic landmarking of syndromic 3D facial surface scans using 2D images. *Sensors*. 2020;20(11):3171. doi:10.3390/s20113171
27. Hammond P, Suttie M. Large-scale objective phenotyping of 3D facial morphology. *Hum Mutat*. 2012;33(5):817-825. doi:10.1002/humu.22054
28. Hostalet N, González A, Salgado-Pineda P, et al. Face-brain correlates as potential sex-specific biomarkers for schizophrenia and bipolar disorder. *Psychiatry Res*. 2024;339:116027. doi:10.1016/j.psychres.2024.116027
29. Halewyck S, Louryan S, Van Der Veken P, et al. Craniofacial embryology and postnatal development of relevant parts of the upper respiratory system. *B-ENT*. 2012;8(Suppl 19):5-11.
30. Marcucio R, Hallgrimsson B, Young NM. Facial morphogenesis: physical and molecular interactions between the brain and the face. *Curr Top Dev Biol*. 2015;115:299-320. doi:10.1016/bs.ctdb.2015.09.001
31. Umeda-Kameyama Y, Kameyama M, Tanaka T, et al. Screening of Alzheimer's disease by facial complexion using artificial intelligence. *Aging (Albany NY)*. 2021;13(2):1765-1772. doi:10.18632/aging.202545
32. Zheng C, Bouazizi M, Ohtsuki T, et al. Detecting dementia from face-related features with automated computational methods. *Bioengineering (Basel)*. 2023;10(7):862. doi:10.3390/bioengineering10070862
33. Lee RW, Chan AS, Grunstein RR, et al. Craniofacial phenotyping in obstructive sleep apnea—a novel quantitative photographic approach. *Sleep*. 2009;32(1):37-45. doi:10.5665/sleep/32.1.37
34. Park JS, Kwon B, Kang HS, et al. Craniofacial phenotype in obstructive sleep apnea and its impact on positive airway pressure (PAP) adherence. *J Pers Med*. 2023;13(8):1196. doi:10.3390/jpm13081196
35. Tepedino M, Illuzzi G, Laurenziello M, et al. Craniofacial morphology in patients with obstructive sleep apnea: cephalometric evaluation. *Braz J Otorhinolaryngol*. 2022;88(2):228-234. doi:10.1016/j.bjorl.2020.05.026
36. Huang L, Xu Y, Gong X, Gao X. Anatomical phenotype of obstructive sleep apnea patients based on cluster analysis. *Orthod Craniofac Res*. 2023;26(4):608-617. doi:10.1111/ocr.12653
37. Ozdemir ST, Ercan I, Can FE, et al. Three-dimensional analysis of craniofacial shape in obstructive sleep apnea syndrome using geometric morphometrics. *Int J Morphol*. 2019;37(1):338-343. doi:10.4067/S0717-95022019000100338
38. Enlow DH, Hans MG. Essentials of facial growth. *Am J Orth Dent Orth*. 1996;110:568.
39. Bulygina E, Mitteroecker P, Aiello L. Ontogeny of facial dimorphism and patterns of individual development within one human population. *Am J Phys Anthropol*. 2006;131(3):432-443. doi:10.1002/ajpa.20317
40. Farrera A, Villanueva M, Vizcaíno A, et al. Ontogeny of the facial phenotypic variability in Mexican patients with 22q11.2 deletion syndrome. *Head Face Med*. 2019;15(1):29. doi:10.1186/s13005-019-0213-9
41. Bookstein FL. *Morphometric Tools for Landmark Data*. Cambridge University Press; 1992. doi:10.1017/CBO9780511573064
42. Dryden I, Mardia K. *Statistical Shape Analysis, with Applications in R: Second Edition*. Wiley; 2016. doi:10.1002/9781119072492
43. Mitteroecker P, Schaefer K. Thirty years of geometric morphometrics: achievements, challenges, and the ongoing quest for biological meaningfulness. *Am J Biol Anthropol*. 2022;178(Suppl 74):181-210. doi:10.1002/ajpa.24531
44. Esteba-Castillo S, Dalmau-Bueno A, Ribas-Vidal N, et al. Adaptacion y validacion del Cambridge Examination for Mental Disorders of Older People with Down's Syndrome and Others with Intellectual Disabilities (CAMDEX-DS) en poblacion española con discapacidad intelectual [Adaptation and validation of CAMDEX-DS (Cambridge Examination for Mental Disorders of Older People with Down's Syndrome and others with intellectual disabilities) in Spanish population with intellectual disabilities]. *Rev Neurol*. 2013;57(8):337-346. doi:10.33588/rn.5708.2013259
45. Montine TJ, Phelps CH, Beach TG, et al. National Institute on Aging - Alzheimer's Association guidelines for the neuropathologic assessment of Alzheimer's disease: a practical approach. *Acta Neuropathol*. 2012;123(1):1-11. doi:10.1007/s00401-011-0910-3
46. Heredia-Lidón A, Echeverry-Quiceno LM, González A, et al. BioFace3D: a fully automatic pipeline for facial biomarkers extraction of 3D face reconstructions segmented from MRI. arXiv 2410.00711 2024. doi:10.48550/arXiv.2410.00711
47. Heredia-Lidón A, García-Mascarell C, Echeverry-Quiceno LM, et al. A critical comparison between template-based and architecture-reused deep learning methods for generic 3D landmarking of anatomical structures. ShapeMI MICCAI Workshop on Shape in Medical Imaging 2024.
48. Paulsen R, Sørensen K, Haspang T, et al. Multi-view consensus CNN for 3D facial landmark placement. *Asian Conference on Computer Vision*. Vol 11361. Springer International Publishing; 2019:706-719. doi:10.1007/978-3-030-20887-5_44
49. Stull KE, Tise ML, Ali Z, Fowler DR. Accuracy and reliability of measurements obtained from computed tomography 3D volume rendered images. *Forensic Sci Int*. 2014;238:133-140. doi:10.1016/j.forsciint.2014.03.005
50. Mitteroecker P, Gunz P. Advances in geometric morphometrics. *Evo Bio*. 2009;36:235-247. doi:10.1007/s11692-009-9055-x
51. Rohlf FJ, Slice D. Extensions of the Procrustes method for the optimal superimposition of landmarks. *Syst Biol*. 1990;39(1):40-59. doi:10.2307/2992207
52. Goodall C. Procrustes methods in the statistical analysis of shape. *J Roy Stat Soc*. 1991;53(2):285-321. doi:10.1111/j.2517-6161.1991.tb01825.x

15306860, 2025, 6, Downloaded from https://faseb.onlinelibrary.wiley.com/doi/10.1096/fj.202403009R by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [25/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

53. Larson JR, Manyama MF, Cole JB, et al. Body size and allometric variation in facial shape in children. *Am J Phys Anthropol*. 2018;165(2):327-342. doi:10.1002/ajpa.23356
54. Adams DC, Otárola-Castillo E. Geomorph: an R package for the collection and analysis of geometric morphometric shape data. *Methods Ecol Evol*. 2013;4(4):393-399. doi:10.1111/2041-210X.12035
55. Campbell NA, Atchley WR. The Geometry of Canonical Variate Analysis. *Syst Zool*. 1981;30(3):268-280. doi:10.2307/2413249
56. Klingenberg CP, Monteiro LR. Distances and directions in multidimensional shape spaces: implications for morphometric applications. *Syst Biol*. 2005;54(4):678-688. doi:10.1080/10635150590947258
57. Klingenberg CP. MorphoJ: an integrated software package for geometric morphometrics. *Mol Ecol Resour*. 2011;11(2):353-357. doi:10.1111/j.1755-0998.2010.02924.x
58. Korayem M, Nuha MA, Waleed B, et al. Craniofacial manifestations of Down syndrome: a review of literature. *Ac J Sci Res*. 2019;7(3):176-181. doi:10.15413/ajsr.2018.0502
59. LaCombe JM, Roper RJ. Skeletal dynamics of Down syndrome: a developing perspective. *Bone*. 2020;133:115215. doi:10.1016/j.bone.2019.115215
60. Thomas JR, Roper RJ. Current analysis of skeletal phenotypes in Down syndrome. *Curr Osteoporos Rep*. 2021;19(3):338-346. doi:10.1007/s11914-021-00674-y
61. González-Agüero A, Vicente-Rodríguez G, Moreno LA, Casajús JA. Bone mass in male and female children and adolescents with Down syndrome. *Osteoporos Int*. 2011;22(7):2151-2157. doi:10.1007/s00198-010-1443-7
62. González-Agüero A, Ara I, Moreno LA, Vicente-Rodríguez G, Casajús JA. Fat and lean masses in youths with Down syndrome: gender differences. *Res Dev Disabil*. 2011;32(5):1685-1693. doi:10.1016/j.ridd.2011.02.023
63. Vioarsdóttir US, O'Higgins P, Stringer C. A geometric morphometric study of regional differences in the ontogeny of the modern human facial skeleton. *J Anat*. 2002;201(3):211-229. doi:10.1046/j.1469-7580.2002.00092.x
64. Allanson JE, O'Hara P, Farkas LG, Nair RC. Anthropometric craniofacial pattern profiles in Down syndrome. *Am J Med Genet*. 1993;47(5):748-752. doi:10.1002/ajmg.1320470530
65. Farkas LG, Katic MJ, Forrest CR. Surface anatomy of the face in Down's syndrome: age-related changes of anthropometric proportion indices in the craniofacial regions. *J Craniofac Surg*. 2002;13(3):368-374. doi:10.1097/00001665-200205000-00002
66. Jayaratne YSN, Elsharkawi I, Macklin EA, et al. The facial morphology in Down syndrome: a 3D comparison of patients with and without obstructive sleep apnea. *Am J Med Genet A*. 2017;173(11):3013-3021. doi:10.1002/ajmg.a.38399
67. Andrade AG, Bubu OM, Varga AW, et al. The Relationship between Obstructive Sleep Apnea and Alzheimer's Disease. *J Alzheimers Dis*. 2018;64(s1):S255-S270. doi:10.3233/JAD-179936
68. Chen CC, Spanò G, Edgin JO. The impact of sleep disruption on executive function in Down syndrome. *Res Dev Disabil*. 2013;34(6):2033-2039. doi:10.1016/j.ridd.2013.03.009
69. Zhang Y, Ren R, Yang L, et al. Sleep in Alzheimer's disease: a systematic review and meta-analysis of polysomnographic findings. *Transl Psychiatry*. 2022;12(1):136. doi:10.1038/s41398-022-01897-y
70. Carmona-Iragui M, Videla L, Lleó A, Fortea J. Down syndrome, Alzheimer disease, and cerebral amyloid angiopathy: the complex triangle of brain amyloidosis. *Dev Neurobiol*. 2019;79(7):716-737. doi:10.1002/dneu.22709
71. McCarron M, McCallion P, Reilly E, et al. A prospective 20-year longitudinal follow-up of dementia in persons with Down syndrome. *J Intellect Disabil Res*. 2017;61(9):843-852. doi:10.1111/jir.12390
72. Zis P, Strydom A. Clinical aspects and biomarkers of Alzheimer's disease in Down syndrome. *Free Radic Biol Med*. 2018;114:3-9. doi:10.1016/j.freeradbiomed.2017.08.024
73. Lao P, Zimmerman ME, Hartley SL, et al. Obstructive sleep apnea, cerebrovascular disease, and amyloid in older adults with Down syndrome across the Alzheimer's continuum. *Sleep Adv*. 2022;3(1):zpac013. doi:10.1093/sleepadvances/zpac013

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Echeverry-Quiceno LM, Llambrich S, Heredia-Lidón Á, et al. Potential of facial biomarkers for Alzheimer's disease and obstructive sleep apnea in Down syndrome and general population. *The FASEB Journal*. 2025;39:e70480. doi:10.1096/fj.202403009R

15306860, 2025, 6, Downloaded from https://faseb.onlinelibrary.wiley.com/doi/10.1096/fj.202403009R by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [25/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License